

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

AIM

Write a C program to simulate a **Deterministic Finite Automata (DFA)** for the language representing strings that **start with 'a' and end with 'a'**.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, len;

    printf("Enter the string: ");
    scanf("%s", str);

    len = strlen(str);

    if(len > 0 && str[0] == 'a' && str[len - 1] == 'a')
        printf("String Accepted\n");
    else
        printf("String Rejected\n");

    return 0;
}
```

Output Clear

Enter the string: aabab
String Rejected

=== Code Execution Successful ===